

DDI assessment for examplinib

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Introduction

This drug-drug interaction (DDI) precipitant risks assessment report is based on the [ICH M12 guidance document](#).

All calculations were done using the [ddir](#) package, version 0.16.0.

Drug properties

The following physico-chemical, ADME and clinical exposure data were used for the calculation of the relevant precipitant drug concentrations:

Table 1: Perpetrator compound parameters for examplinib

parameter	value	source
oral	1	
MW (g/mol)	420.5	
$dose$ (mg)	50	clinical dose
$solubility$ (mg/l)	Inf	
$C_{max,ss}$ (ng/ml)	566	study 001
f_u	0.046	study 002
$f_{u,mic}$	1	
R_B	0.9	study 003
F_a	1	
F_g	1	
k_a (1/min)	0.1	

Key perpetrator concentrations

The following perpetrator concentrations were calculated as described in Appendix 1:

Table 2: Key perpetrator concentrations for exemplinib

parameter	value (ng/ml)	value (μM)
I_{gut}	200000.00	475.62
$I_{max,ss,u}$	26.04	0.06
$I_{max,inlet,u}$	184.18	0.44
$I_{max,intestinal}$	16666.67	39.64

DDI risk as inhibitor or inducer of drug-metabolizing enzymes

Basic modeling of CYP inhibition

Reversible inhibition

Following the basic modeling approach (refer to Section 2.1.2.1 of the [ICH M12 guideline](#)), the relevant metric for the assessment of the direct CYP inhibition risk is $R = [I]/K_{i,u}$ with the inhibitor concentration $[I]$ being $C_{max,ss,u}$ for hepatic CYP enzymes and I_{gut} for intestinal CYP enzymes.

R values of $R \geq 0.02$ and $R \geq 10$ for hepatic and intestinal enzymes are considered to indicate a potential clinical risk.

Table 3: Direct CYP inhibition risk for exemplinib

tar- get	K_i (μM)	$K_{i,u}$ (μM)	source	risk R (hepatic)		risk R_{gut} (intestinal)	
CYP2C9	7.01	7.01	study 001	0.01	No	NA	
CYP2C19	48.70	48.70	study 001	0.00	No	NA	
CYP3A4	100.00	100.00	study 001	0.00	No	4.76	No

Time-dependent inhibition

The risk for time-dependent inhibition (TDI) of CYP enzymes is assessed based on the formula given in Section ‘[Time-dependent CYP inhibition](#)’ in Appendix 1, where $R \geq 1.25$ suggest a clinically relevant DDI potential that requires further investigation (refer to Section 2.1.2.2 of the [ICH M12 guideline](#)).

Table 4: Time-dependent CYP inhibition risk for exemplinib

target	K_i (μM)	f_u	k_{inact} (1/h)	k_{deg} (1/h)	source	R	risk
CYP3A4	0.17	0.046	0.04	0.0193	study 001	2.34	Yes

Modeling of CYP induction

Basic mRNA ‘fold-change’ method

The basic mRNA ‘fold-change’ approach evaluates whether the maximal change in CYP mRNA expression is ≥ 2 -fold at concentrations up to 50-fold above the expected unbound systemic concentration of the drug (refer to Section 2.1.4.1 of the [ICH M12 guideline document](#)).

Basic modeling as per the ICH M12 guideline results in the following risk assessment. Data from CYP induction studies that did not cover a concentration range up to 50-fold over $C_{max,ss,u}$ are flagged in the table. In these cases, CYP mRNA fold-changes (E_{max}) < 2 -fold cannot fully exclude a CYP induction risk.

Table 5: Static CYP induction risk for exemplinib

target	E_{max}	$maxc$ (μM)	source	risk	note
CYP1A2	1.00	5	study 007	No	
CYP2B6	1.00	5	study 007	No	
CYP2C8	NA	NA	NA		
CYP2C9	NA	NA	NA		
CYP2C19	NA	NA	NA		
CYP2D6	NA	NA	NA		
CYP3A4	7.35	3	study 007	Yes	Not tested up to 50-fold $C_{max,u}$

Basic kinetic method

The basic kinetic method for the assessment of CYP induction is based on the EC_{50} and E_{max} parameters derived from in vitro studies (refer to Section ‘[Basic kinetic modeling of CYP induction](#)’ in Appendix 1). For $R < 0.8$, an in vivo induction risk cannot be excluded:

Table 6: Kinetic CYP induction risk for exemplinib

target	E_{max}	EC_{50}	$maxc$ (μM)	source	R	risk
CYP1A2	1.00	NA	5	study 007	NA	
CYP2B6	1.00	NA	5	study 007	NA	
CYP2C8	NA	NA	NA	NA	NA	
CYP2C9	NA	NA	NA	NA	NA	
CYP2C19	NA	NA	NA	NA	NA	
CYP2D6	NA	NA	NA	NA	NA	
CYP3A4	7.35	1.64	3	study 007	0.33	Yes

Mechanistic static modeling

Using the mechanistic static modeling approach (refer to Section 7.5.1.2 of the [ICH M12 guideline](#), AUC ratios for specific sensitive CYP substrates are calculated, considering the available in vitro data for both direct and time-dependent inhibition, and mRNA induction (refer to Section ‘[Mechanistic static modeling of CYP modulation](#)’ in Appendix 1).

Mechanistic static modeling may be used to investigate CYP inhibition alone, or both inhibition and induction effects. AUC ratios outside the 0.8 to 1.25 interval are considered to indicate a clinical risk.

CYP inhibition only

Table 7: Mechanistic-static CYP inhibition risk for exemplinib

tar- get	sub- strate	$K_{i,u}$ (μM)	f_{gut}	fm	fm_{CYP}	Ag	Ah	Bg	Bh	Cg	Ch	R_{AUC}	risk
CYP2C9	warfarin	7.01	1.00	1.00	0.91	0.15	0.94	1.00	1.0	1	1	1.06	No
CYP2C19	clopidogrel	48.70	1.00	1.00	0.87	0.55	0.99	1.00	1.0	1	1	1.01	No
CYP3A4	midazolam	100.00	0.57	0.96	1.00	0.72	1.00	0.43	0.4	1	1	3.36	Yes

CYP inhibition and induction

Table 8: Mechanistic-static CYP inhibition risk for examplinib

tar- get	sub- strate	$K_{i,u}$ (μM)	f_{gut}	fm	fm_{CYP}	Ag	Ah	Bg	Bh	Cg	Ch	R_{AUC}	risk
CYP2C9	warfarin	7.01	1.00	1.00	0.91	0.15	0.94	1.00	1.0	1.00	1.00	1.06	No
CYP2C19	clopidogrel	48.70	1.00	1.00	0.87	0.55	0.99	1.00	1.0	1.00	1.00	1.01	No
CYP3A4	midazolam	100.00	0.57	0.96	1.00	0.72	1.00	0.43	0.4	8.06	2.55	0.60	Yes

Basic modeling of UGT inhibition

The relevant metric for basic modeling of the UGT inhibition risk is $R = C_{max,ss,u}/K_{i,u}$ (refer to Section 2.1.2.1 of the [ICH M12 guidance document](#)) for details.

$R > 0.02$ is considered to indicate a potential UGT inhibition risk.

Note that in in vitro UGT inhibition studies, often IC_{50} rather than K_i values are reported. Assuming that substrate concentrations close to K_m are used, K_i is calculated as $K_i = IC_{50}/2$ (refer to Cheng, Prusoff 1973).

Basic modeling of UGT inhibition results in the following risk assessment:

Table 9: UGT inhibition risk for examplinib

target	K_i (μM)	$K_{i,u}$ (μM)	source	R	risk
UGT1A1	15.0	15.0	study 009	0.00	No
UGT1A3	15.0	15.0	study 009	0.00	No
UGT1A4	15.0	15.0	study 009	0.00	No
UGT1A6	15.0	15.0	study 009	0.00	No
UGT1A9	3.8	3.8	study 009	0.02	No
UGT2B7	15.0	15.0	study 009	0.00	No
UGT2B15	15.0	15.0	study 009	0.00	No
UGT2B17	6.1	6.1	study 009	0.01	No

DDI risk as inhibitor of drug transporters

The metric for the assessment of the drug transporter inhibition risk is $R = [I]/IC_{50,u}$. For the perpetrator concentrations relevant for the individual transporters, refer to Section ‘[Inhibition of drug transporters](#)’ in Appendix 1.

Note that OCT1 is currently not covered by the M12 guideline.

Table 10: Transporter inhibition risk for examplnib

target	IC_{50}	source	i	R	threshold	risk
Pgp_int	0.41	study 005	igut	1160.06	10.00	Yes
Pgp_sys	0.41	study 005	imaxssu	0.15	0.02	Yes
BCRP_int	1.90	study 005	igut	250.33	10.00	Yes
BCRP_sys	1.90	study 005	imaxssu	0.03	0.02	Yes
OATP1B1	177.00	study 006	imaxinletu	0.00	0.10	No
OATP1B3	35.00	study 006	imaxinletu	0.01	0.10	No
OAT1	271.00	NA	imaxssu	0.00	0.10	No
OAT3	300.00	NA	imaxssu	0.00	0.10	No
BSEP	12.80	NA	imaxssu	0.00	0.10	No
OCT1	2.30	study 006	NA	NA	NA	
OCT2	67.00	study 006	imaxssu	0.00	0.10	No
MATE1	3.60	study 006	imaxssu	0.02	0.02	No
MATE2k	1.10	study 006	imaxssu	0.06	0.02	Yes

Appendix 1: Calculations and formulae

Relevant precipitant drug concentrations

Gut concentration

The maximal gut concentration (I_{gut}) for the orally administered compounds is the administered dose dissolved in 250 ml.

$$I_{gut} = \frac{D}{250}$$

Systemic concentration

The unbound systemic ($C_{max,ss,u}$) concentration is considered the relevant precipitant concentration for hepatic enzyme inhibition and induction:

$$C_{max,ss,u} = I_{max,ss} * f_u$$

Hepatic inlet concentration

The hepatic inlet concentration is considered the relevant perpetrator concentration for inhibition of the hepatic uptake transporters OATP1B1 and OATP1B3, and for the hepatic terms in the mechanistic static modeling equation (refer to Section ‘[Mechanistic static modeling of CYP inhibition/induction]’).

The hepatic inlet concentration is composed of the systemic concentration and the portal contribution. For orally administered drugs, the portal term is calculated as:

$$portal\ term = D * \frac{F_a * F_g * k_a}{Q_h * R_B} * 1000\ ng/ml$$

with

- D the administered dose in mg
- F_a the fraction absorbed after oral administration
- F_g the fraction available after gut metabolism
- k_a the absorption rate
- Q_h the hepatic blood flow
- R_B the blood-to-plasma ratio.

The standard hepatic blood flow is assumed as 97 l/h/70 kg or 1.61 l/min/70 kg.

The relevant hepatic inlet ($I_{max,inlet,u}$, also called I_h in the mechanistic static modeling equations) concentration is the sum of the maximal systemic plasma concentration and the portal contribution:

$$I_{max,inlet,u} = (C_{max,ss} + portal\ term) * f_u$$

Enteric concentration

For the parent compound, the villous concentration in the gut ($I_{enteric}$, also called I_g in the mechanistic static modeling equations) is calculated as:

$$I_{enteric,u} = D * \frac{F_a * k_a}{Q_{ent}} * 1000\ ng/ml$$

with

- F_a the fraction absorbed after oral administration
- k_a the absorption rate constant
- Q_{ent} the enteric villous blood flow

Note that as per the ICH M12 guideline and [Rostami-Hodjegan and Tucker, 2004](#) the blood-to-plasma distribution ratio and the plasma binding of the drug are not applicable for the calculation of the villous concentration.

The standard villous blood flow is assumed as 18 l/h/70 kg or 0.3 l/min/70 kg.

Basic modeling of enzyme inhibition

Reversible inhibition

For the basic modeling of direct (reversible) enzyme inhibition, the ratios of the relevant inhibitor concentration to the $K_{i,u}$ are considered (refer to Section 2.1.2.1 of the [ICH M12 guidance document](#)).

For in vitro studies conducted using human liver microsomes, the microsomal unbound fraction, $f_{u,mic}$ is used to calculate $K_{i,u}$. If unknown, a default of 1 is assumed.

R values larger than 0.02 (liver) or 10 (gut), are considered to indicate a potential clinical enzyme inhibition risk using this method.

Liver

$$R = \frac{C_{max,ss,u}}{K_{i,u}}$$

Gut wall

$$R_{gut} = \frac{I_{gut}}{K_{i,u}}$$

Time-dependent CYP inhibition

For the basic modeling of the potential for time-dependent CYP inhibition (TDI), the following metric is considered:

$$R = \frac{k_{obs} + k_{deg}}{k_{deg}}$$

with

$$k_{obs} = \frac{5 * k_{inact} * C_{max,u}}{K_{I,u} + 5 * C_{max,u}}$$

The CYP degradation constant, k_{deg} is a physiological constant that should be derived from the scientific literature. In this DDI assessment report, standard values are used unless otherwise indicated.

Values of $R \geq 1.25$ is considered to indicate a clinically relevant TDI potential and suggest the need for further investigation.

Basic mRNA fold-change method method to assess CYP induction

This basic risk assessment evaluates the mRNA induction for a set of hepatocyte batches from different donors. Increases of CYP enzyme mRNA ≥ 2 -fold at concentrations up to 50-fold above $C_{max,u}$ is considered to indicate a clinical risk for CYP induction.

In the context of this assessment only the worst-case donor data is considered.

Basic kinetic modeling of CYP induction

For the basic kinetic modeling of the CYP induction potential, the following metric is considered (refer to Section 2.1.4.3 of the ICH M12 guideline):

$$R = \frac{1}{1 + d * \frac{E_{max} * 10 * C_{max,ss,u}}{EC_{50,u} + 10 * C_{max,ss,u}}}$$

with d a scaling factor that has a standard value of 1. A different value can be used if warranted by prior experience with the experimental conditions.

$R \leq 0.8$ suggest a relevant in vivo CYP induction potential.

Mechanistic static modeling of CYP modulation

In this approach, AUC ratios for specific DDI object substrates are projected based on their known intestinal and hepatic metabolism. Both direct (competitive) and time-dependent inhibition, as well as enzyme induction are considered. AUC ratios are calculated according to the below formula (refer to Section 7.5.1.2 of the ICH M12 guideline):

$$AUCR = \frac{1}{A_g * B_g * C_g * (1 - F_g) + F_g} * \frac{1}{A_h * B_h * C_h * f_m + (1 - f_m)}$$

This calculation is applied for typical probe substrates for which F_g , i.e., the fraction escaping gut metabolism and f_m , i.e., the fraction metabolized are known.

Note that the f_m is composed of the overall fraction metabolized for the respective probe substrate, and the fraction metabolized by the CYP enzyme in questions:

$$f_m = f_{m,overall} * f_{m,CYP}$$

The individual terms in the AUC calculation are:

Reversible inhibition

$$A_g = \frac{1}{1 + \frac{I_g}{K_i}}$$

$$A_h = \frac{1}{1 + \frac{I_h}{K_i}}$$

Time-dependent inhibition

$$B_g = \frac{k_{deg,g}}{k_{deg,g} + \frac{I_g * k_{inact}}{I_g + K_I}}$$

$$B_h = \frac{k_{deg,h}}{k_{deg,h} + \frac{I_h * k_{inact}}{I_h + K_I}}$$

Induction

$$C_g = 1 + \frac{d * E_{max} * I_g}{I_g + EC_{50}}$$

$$C_h = 1 + \frac{d * E_{max} * I_h}{I_h + EC_{50}}$$

with the hepatic inlet concentration $I_h = I_{max,inlet,u}$ and the intestinal concentration $I_g = I_{enteric}$ (see above). d is an induction scaling factor (assumed to be 1 but can be adjusted based on the experimental conditions).

If the predicted AUC ratio is outside of the 0.8 to 1.25 interval, further evaluation is required.

Inhibition of drug transporters

As per the M12 guideline, the metric for the assessment of the drug transporter inhibition risk is:

$$R = [I]/IC_{50,u}$$

In the respective in vitro studies, the substrate concentration is usually very low, so that $K_i \approx IC_{50}$ can be assumed. Under common assay conditions, no protein is added to the medium so that the fraction unbound can be assumed 1, i.e. $IC_{50} = IC_{50,u}$.

The following relevant participant concentrations $[I]$ and regulatory thresholds of concern apply for the transporters:

I	transporter	threshold
I_{gut}	P-gp and BCRP when drugs are orally administered	10

I	transporter	threshold
$C_{max,ss,u}$	P-gp and BCRP when drugs are administered parenterally or for drug metabolites	0.02
$I_{max,inlet,u}$	hepatic basolateral transporters OCT1, OATP1B1 and OATP1B3	0.1
$C_{max,ss,u}$	renal basolateral transporters OAT1, OAT3 and OCT2	0.1
$C_{max,ss,u}$	apical transporters MATE1 and MATE2-K	0.02

Refer to Section ‘[Relevant precipitant drug concentrations](#)’ for the calculation of the relevant precipitant concentrations.

Appendix 2: R Session Info

This document was created using R version 4.5.2 (2025-10-31) and the following packages:

name	version
ddir	0.16.0
knitr	1.51
lubridate	1.9.5
forcats	1.0.1
stringr	1.6.0
dplyr	1.2.0
purrr	1.2.1
readr	2.1.6
tidyr	1.3.2
tibble	3.3.1
ggplot2	4.0.2
tidyverse	2.0.0